IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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21 October 2026

GitHub: https://github.com/martimramos/ibmds



Executive Summary

Summary of methodologies

- Data collection: SpaceX REST API + Wikipedia web scraping → 90 Falcon 9 launches (2010–2020)
- Data wrangling: landing outcome → binary label Class (1 = landed); one-hot encoding → 80 features
- EDA with visualization (Seaborn) and SQL (10 queries on SQLite)
- Interactive visual analytics: Folium maps and a Plotly Dash dashboard
- Predictive analysis: Logistic Regression, SVM, Decision Tree, KNN tuned with GridSearchCV (cv = 10)

Summary of all results

- EDA: 66.7% of launches landed; success rose from 0% (2010–13) to 90% (2019) and grows with flight number
- SQL: 4 launch sites; NASA (CRS) payload 45,596 kg; first ground-pad landing 22 Dec 2015; 100 of 101 missions succeeded
- Folium: all sites are coastal; KSC LC-39A has the best success ratio; CCAFS SLC-40 is 0.49 km from the coast
- Dash: KSC LC-39A has 41.7% of all successes and a 76.9% success ratio; 2,000–4,000 kg payloads land most often
- Classification: LogReg, SVM and KNN reach 83.3% test accuracy; SVM is best (84.8% CV), Decision Tree 72.2%

GitHub repository (all notebooks + Dash app): https://github.com/martimramos/ibmds

Introduction

Project background and context

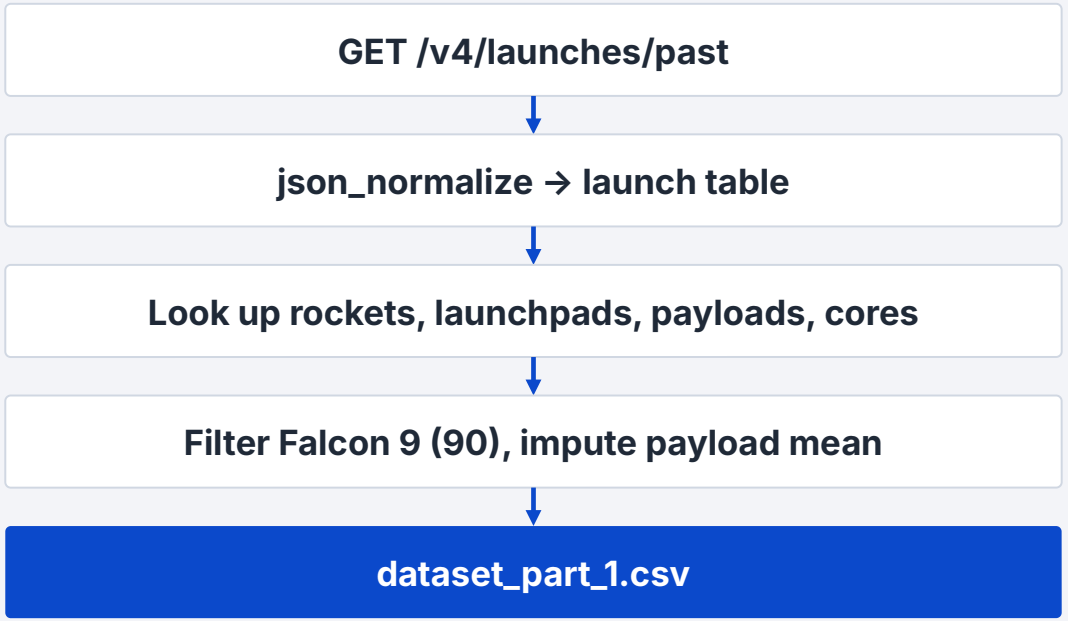
- SpaceX advertises Falcon 9 launches at about $62 M; other providers charge upwards of $165 M
- Most of the saving comes from landing and reusing the first stage
- If we can predict whether the first stage will land, we can estimate the cost of a launch, which is useful for a competitor such as Space Y bidding against SpaceX

Problems we want to find answers to

1. Which factors (site, payload, orbit, booster, flight history) drive a successful landing?
2. How has landing success changed over time?
3. Where are the launch sites, and what surrounds them?
4. Can a classifier predict the landing outcome, and how reliably?

Data Collection Methodology: SpaceX API and Web Scraping

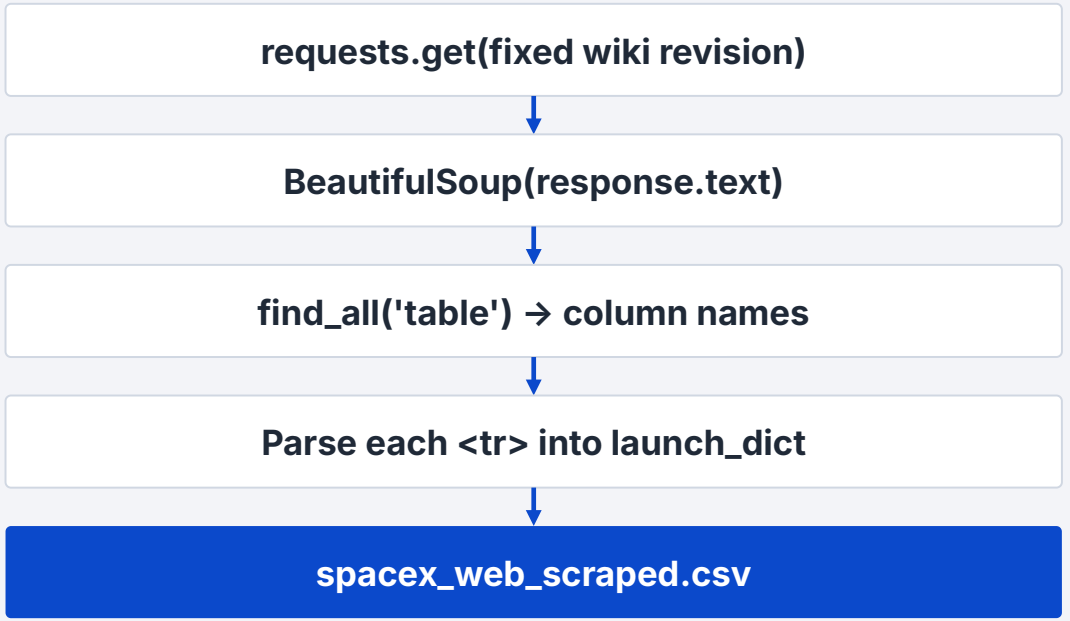
SpaceX REST API



Single-core, single-payload launches up to 13 Nov 2020; 26 missing LandingPad values kept

GitHub lab URL:
https://github.com/martimramos/ibmds/blob/main/01_spacex_data_collection_api.ipynb

Web scraping (Wikipedia)



"List of Falcon 9 and Falcon Heavy launches": flight, date, booster, site, payload, mass, orbit, customer, outcome

GitHub lab URL:
https://github.com/martimramos/ibmds/blob/main/02_spacex_webscraping.ipynb

Data Wrangling Methodology

- Checked missing values and data types per column
- Counted launches per site, per orbit and per landing outcome
- Outcome = mission result + landing type (ASDS = drone ship, RTLS = ground pad, Ocean)
- Defined bad_outcomes = landing 'False' or 'None' → Class 0; otherwise Class 1
- Overall success rate: 66.7% → saved as dataset_part_2.csv
- For modelling: one-hot encoded Orbit, LaunchSite, LandingPad, Serial → 80 numeric features (dataset_part_3.csv)

Outcome	Launches	Class
True ASDS	41	1
None None	19	0
True RTLS	14	1
False ASDS	6	0
True Ocean	5	1
False Ocean	2	0
None ASDS	2	0
False RTLS	1	0

GitHub lab URL: https://github.com/martimramos/ibmds/blob/main/03_spacex_data_wrangling.ipynb

EDA Methodology: Data Visualization and SQL

Charts plotted, and why

Chart	Purpose
Flight Number vs. Launch Site (scatter)	experience effect per site
Payload vs. Launch Site (scatter)	payload limits per site
Success rate by Orbit (bar)	compare mission profiles
Flight Number vs. Orbit (scatter)	experience effect per orbit
Payload vs. Orbit (scatter)	payload effect per orbit
Yearly success rate (line)	trend over time

SQL queries performed (SQLite, %sql)

- Distinct launch site names
- 5 records where site begins with ‘CCA’
- Total payload mass for NASA (CRS)
- Average payload mass of F9 v1.1
- First successful ground-pad landing date
- Drone-ship successes with 4,000–6,000 kg payload
- Count of mission outcomes
- Boosters with the maximum payload (sub-query)
- 2015 drone-ship failures (month, booster, site)
- Rank landing outcomes 2010-06-04 to 2017-03-20

GitHub lab URL: https://github.com/martimramos/ibmds/blob/main/05_spacex_eda_dataviz.ipynb
GitHub lab URL: https://github.com/martimramos/ibmds/blob/main/04_spacex_eda_sql.ipynb

Interactive Visual Analytics Methodology: Folium and Plotly Dash

Folium map objects

Circle + Marker with DivIcon label for each launch site, to locate all sites

MarkerCluster of all 56 launch records, to group launches per site

Green / red Icons for landed / failed, to read site success at a glance

MousePosition to read coordinates of coast, roads, rail and cities

PolyLine + distance label (Haversine), to measure proximity to infrastructure

Plotly Dash components

Dropdown (All Sites or one site): drives both charts

Pie chart: share of successes by site, or success vs. failure for one site, to compare and drill down

Range slider (payload 0–10,000 kg), to isolate payload bands

Scatter plot: payload vs. outcome coloured by booster version, to spot payload and booster effects

Two callbacks connect the inputs to the charts

GitHub lab URL: https://github.com/martimramos/ibmds/blob/main/06_spacex_launch_site_location_folium.ipynb
GitHub lab URL: https://github.com/martimramos/ibmds/blob/main/07_spacex_dash_app.py

Predictive Analysis Methodology (Classification)



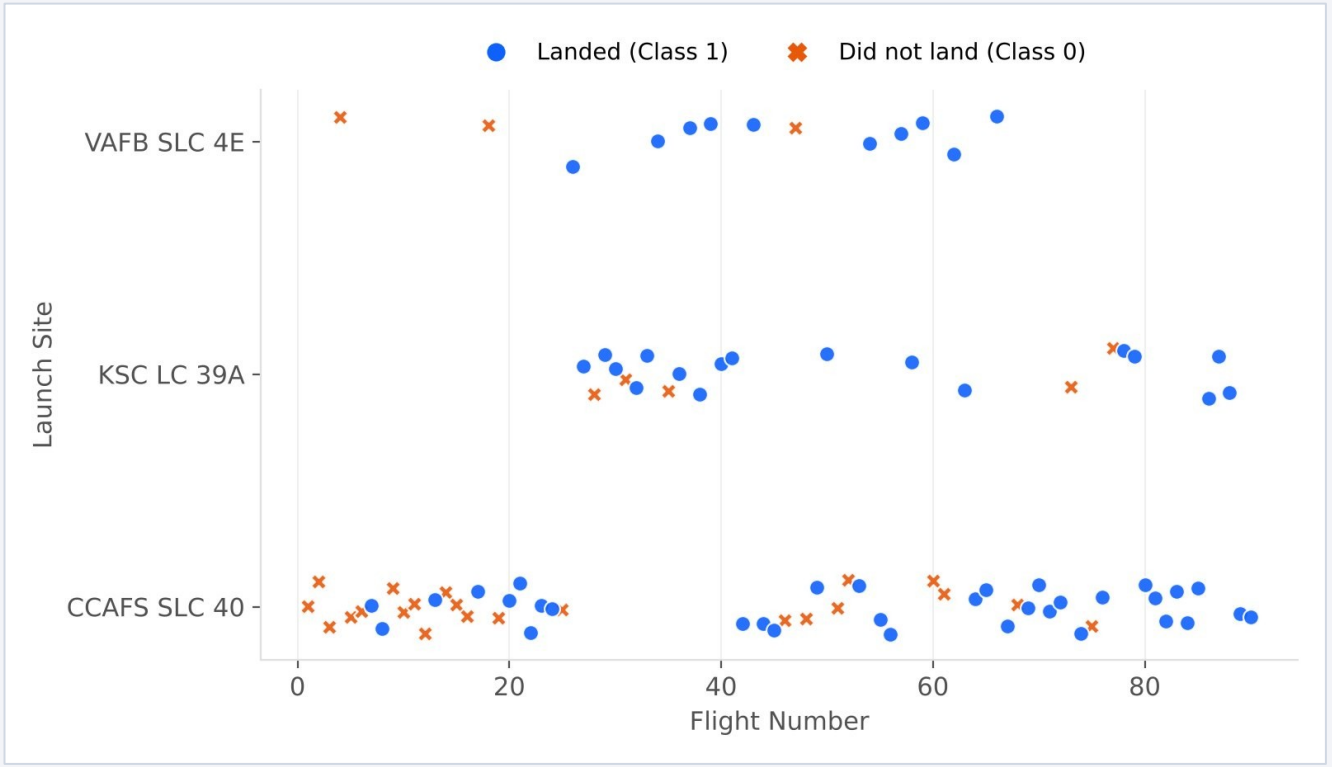
Model	Hyper-parameter grid	Best parameters
Logistic Regression	C ∈ {0.01, 0.1, 1}, l2, lbfgs	C = 1
SVM	kernel ∈ {linear, rbf, poly, sigmoid}, C, γ ∈ logspace(-3, 3, 5)	sigmoid, C = 1, γ = 0.032
Decision Tree	gini/entropy, best/random, depth 2–18, leaf, split	gini, depth 10
KNN	k = 1–10, algorithm, p ∈ {1, 2}	k = 6, p = 1 (Manhattan)

Built 4 models, evaluated each on the held-out test set, improved the choice by re-scoring with 20× repeated 5-fold CV, and selected the model with the best accuracy.

GitHub lab URL: https://github.com/martimramos/ibmds/blob/main/08_spacex_machine_learning_prediction.ipynb

EDA with Visualization: Flight Number vs. Launch Site

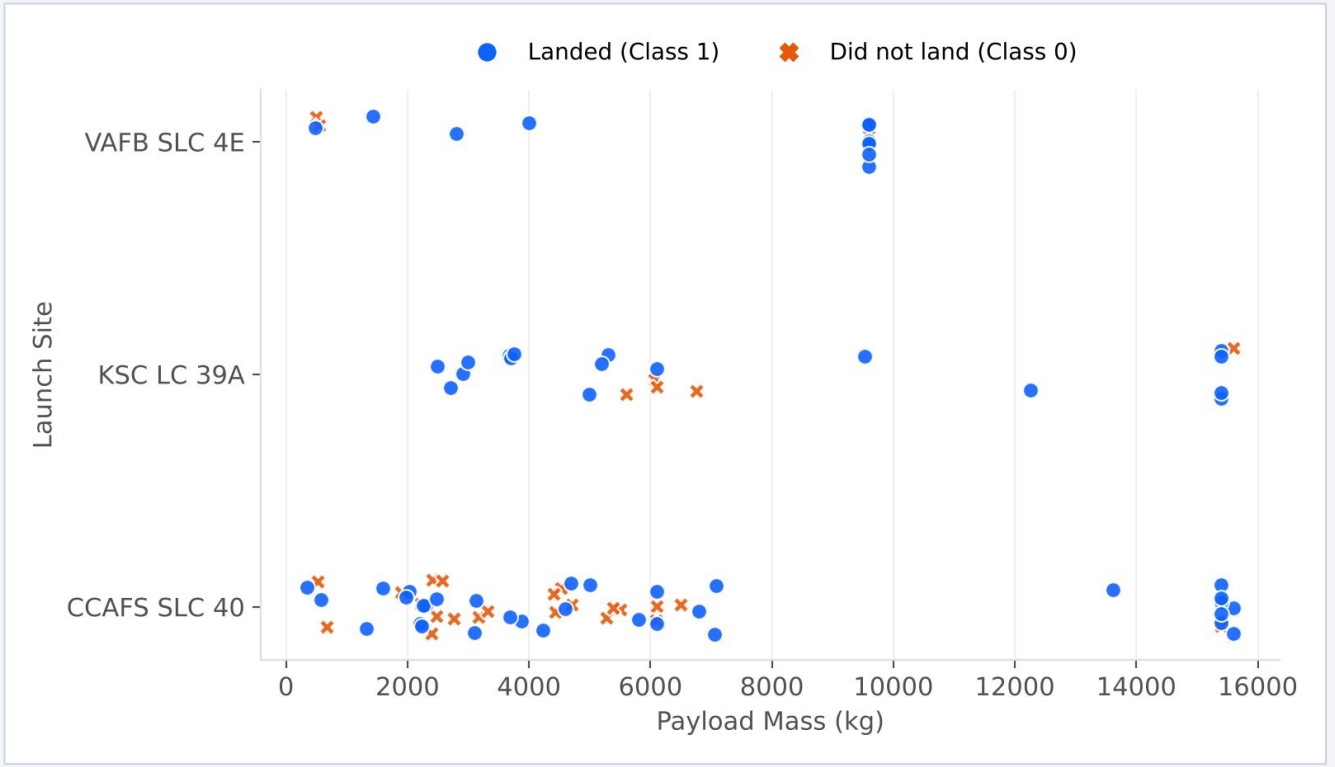
- Scatter plot of flight number vs. launch site, coloured by outcome
- CCAFS SLC 40 hosts 55 of 90 launches, including most early failures
- VAFB SLC 4E and KSC LC 39A came online after ~flight 20
- At every site later flights are mostly successes



Key insight Landing success grows with flight number at all sites; the site effect is partly an experience effect.

EDA with Visualization: Payload vs. Launch Site

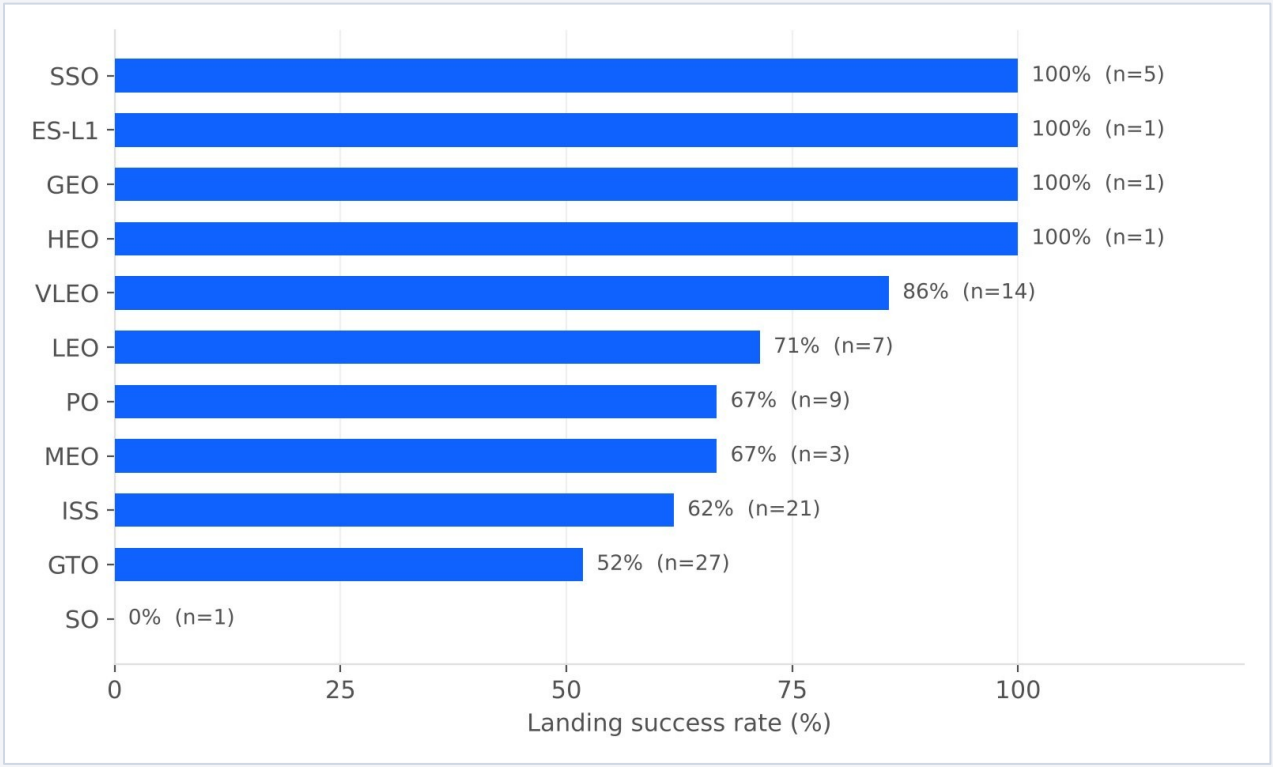
- Scatter plot of payload mass vs. launch site
- VAFB SLC 4E launched no payload above 10,000 kg
- Heavy payloads (> 12,000 kg) flew from CCAFS and KSC and nearly all landed



Key insight Site success rates: KSC 77%, VAFB 77%, CCAFS 60%. Payload alone does not explain failures.

EDA with Visualization: Success Rate vs. Orbit Type

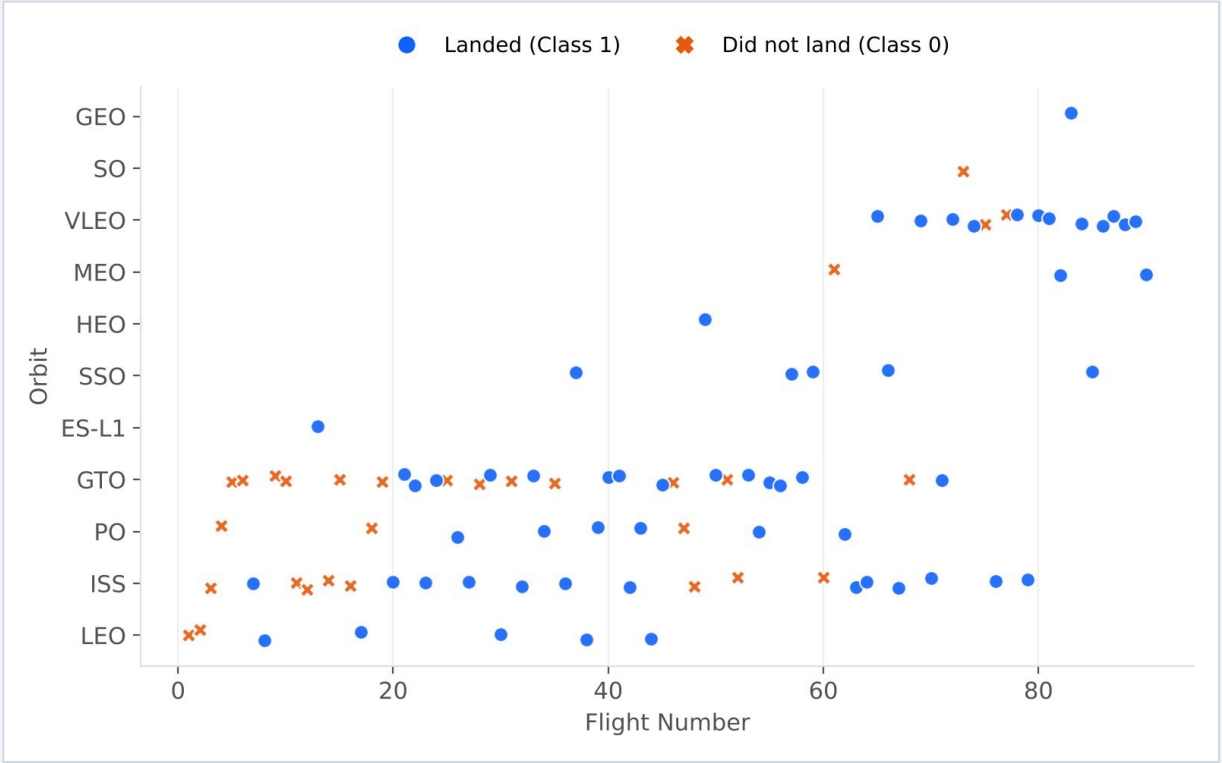
- Bar chart of landing success rate for each orbit
- ES-L1, GEO, HEO, SSO: 100% (n ≤ 5)
- VLEO 86%, LEO 71%, PO 67%, ISS 62%, GTO 52%
- SO: 0% (one launch)



Key insight GTO (27 launches) and ISS (21) are the real hard cases; the 100% orbits have very few launches.

EDA with Visualization: Flight Number vs. Orbit Type

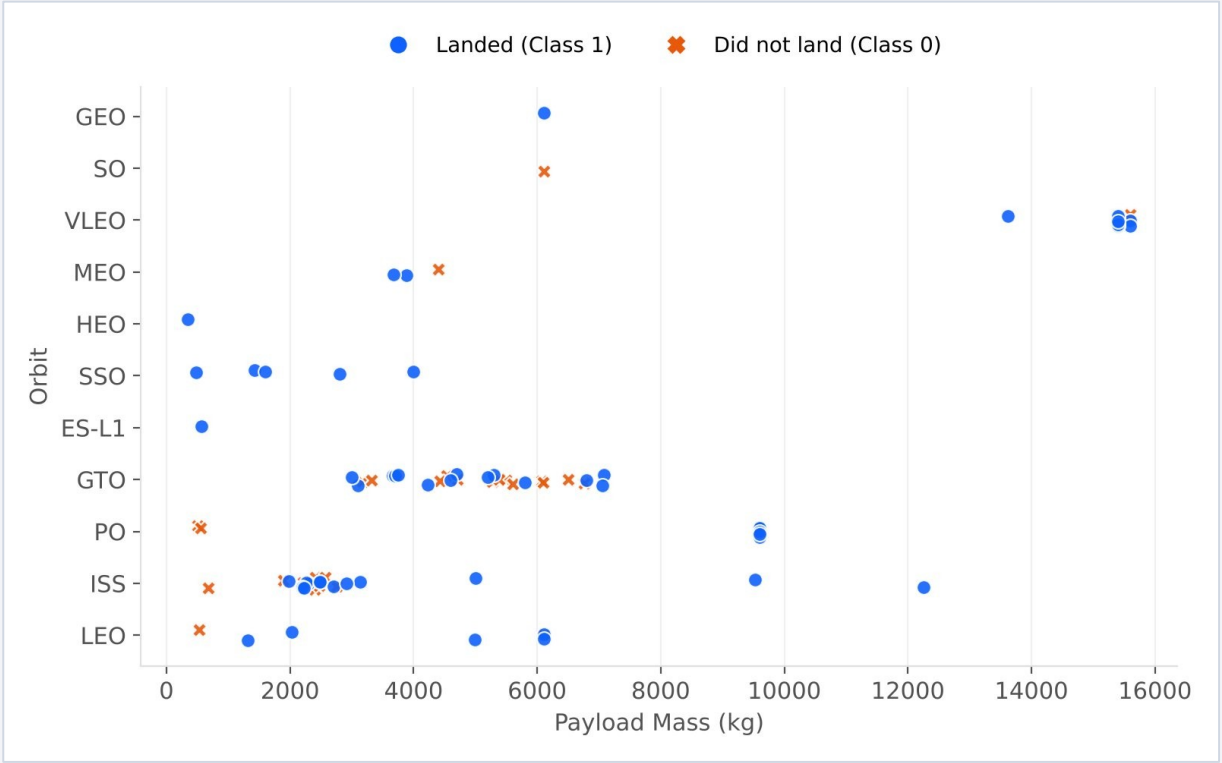
- Scatter plot of flight number vs. orbit
- LEO: success appears related to the number of flights
- GTO: no clear relationship with flight number
- VLEO starts after flight ~60 and mostly lands



Key insight Newer orbit types (VLEO, SSO) were flown once SpaceX was experienced, which inflates their success rates.

EDA with Visualization: Payload vs. Orbit Type

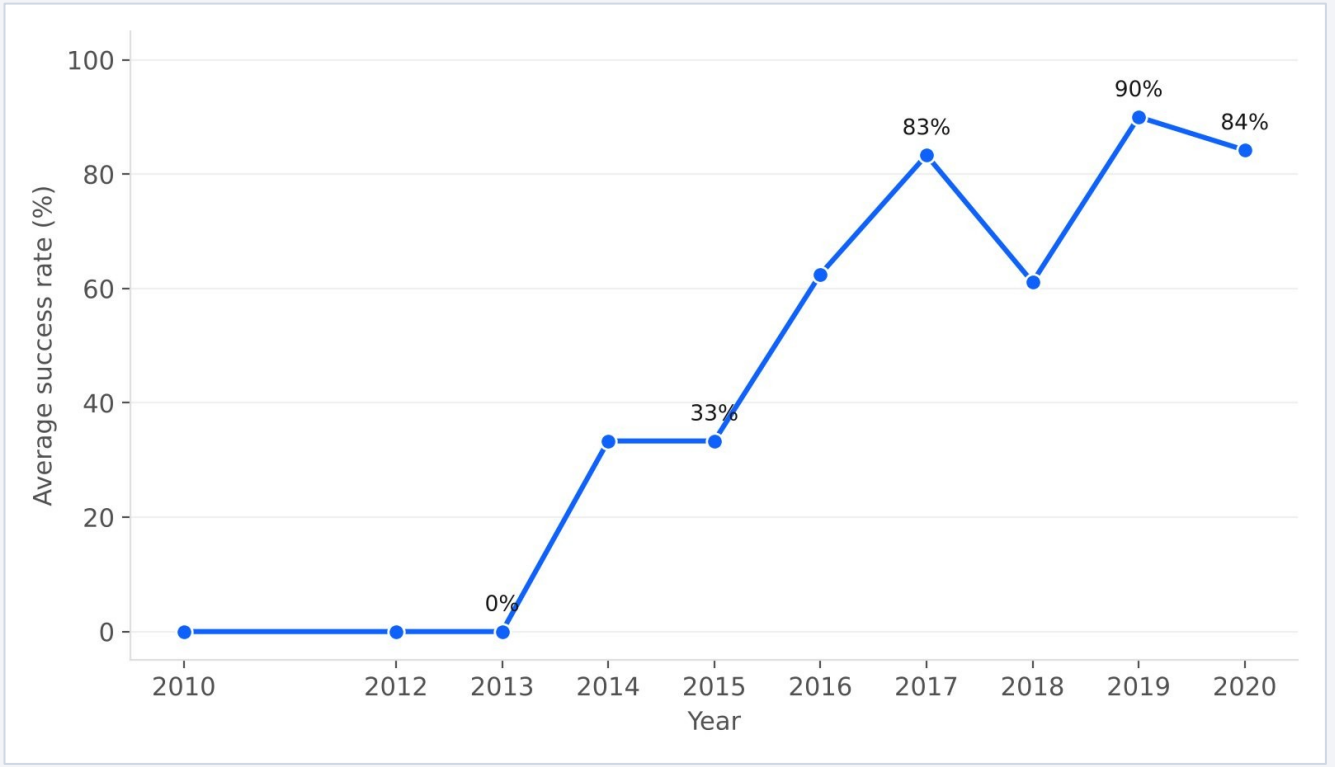
- Scatter plot of payload mass vs. orbit
- Heavy payloads land more often for Polar, LEO and ISS orbits
- GTO mixes successes and failures at every payload level
- VLEO carries the heaviest payloads (15,600 kg)



Key insight Payload interacts with orbit: heavy is fine for LEO-type missions, while GTO stays uncertain at any mass.

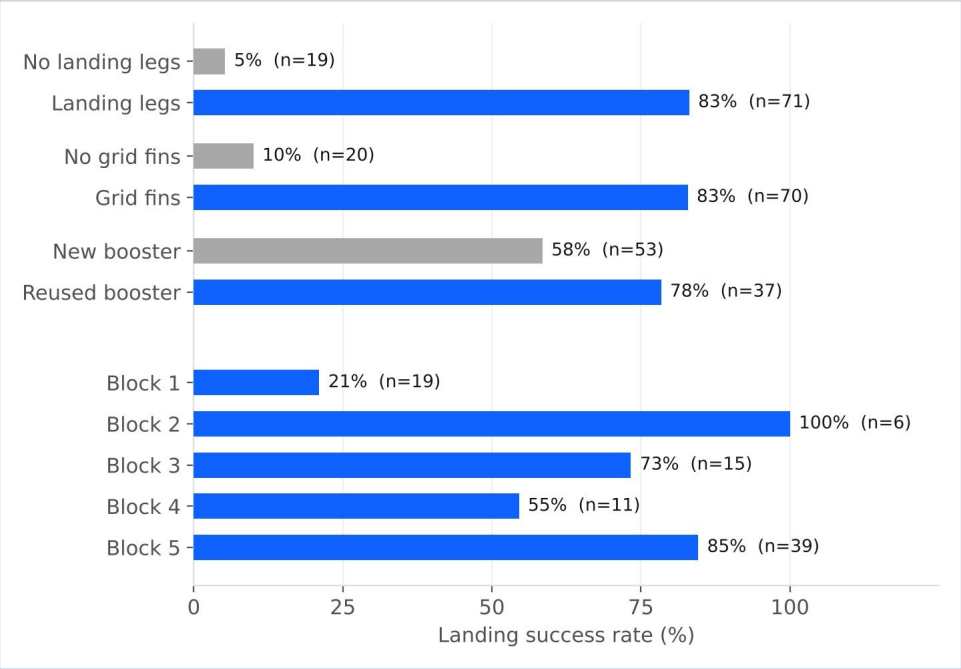
EDA with Visualization: Launch Success Yearly Trend

- Line chart of the average success rate per year
- 2010–2013: 0%
- 2014–2015: 33%; 2016: 62%
- 2017: 83%, dip in 2018 (61%)
- 2019: 90%; 2020: 84%

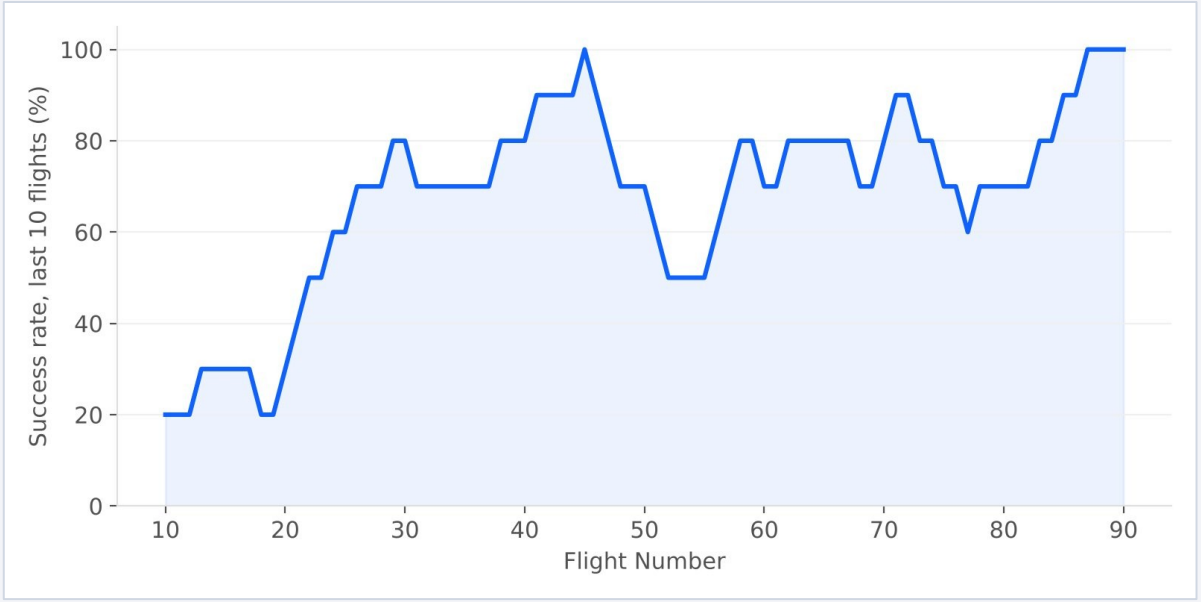


Key insight The success rate has kept increasing since 2013, which is consistent with a strong learning curve.

Innovative Insights: Hardware and the Learning Curve



Launches without legs / grid fins were not landing attempts. Among the 71 real attempts, 83% landed; reused boosters 78% vs. new 58%; Block 5 85%



10-flight rolling success rate: flights 1–30 43%, 31–60 73%, 61–90 83%. It first reached 80% at flight 29 and ended at 100%

Key insight For a competitor, the key question is “will SpaceX attempt a landing?”. When it does, it usually succeeds, and recent flights matter most.

EDA with SQL: Launch Sites and ‘CCA’ Records

Q1 **Find the names of the unique launch sites**

```
SELECT DISTINCT Launch_Site
FROM SPACEXTABLE
```

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

Q2 **Find 5 records where launch sites begin with ‘CCA’**

```
SELECT * FROM SPACEXTABLE
WHERE Launch_Site LIKE 'CCA%'
LIMIT 5
```

Result (5 rows) shown below: all are F9 v1.0 flights from CCAFS LC-40, 2010–2013.

Date	Booster	Site	Payload	kg	Landing_Outcome
2010-06-04	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	Failure (parachute)
2010-12-08	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, b	0	Failure (parachute)
2012-05-22	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	No attempt
2012-10-08	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	No attempt
2013-03-01	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	No attempt

Key insight 4 launch sites (3 Florida, 1 California). The first CCA records are early Dragon demo and NASA CRS flights.

EDA with SQL: Payload Totals, Averages and First Landing

Q3 Total payload mass carried by boosters for NASA (CRS)

```
SELECT SUM(PAYLOAD_MASS__KG_) AS Total_Payload_kg
FROM SPACEXTABLE
WHERE Customer = 'NASA (CRS)'
```

45,596 kg

NASA's CRS resupply missions: a key recurring customer

Q4 Average payload mass carried by booster version F9 v1.1

```
SELECT AVG(PAYLOAD_MASS__KG_) AS Avg_Payload_kg
FROM SPACEXTABLE
WHERE Booster_Version = 'F9 v1.1'
```

2,928.4 kg

(2,534.7 kg if LIKE 'F9 v1.1%' includes all v1.1 serials)

Q5 Date of the first successful landing on a ground pad

```
SELECT MIN(Date) AS First_Ground_Pad_Success
FROM SPACEXTABLE
WHERE Landing_Outcome = 'Success (ground pad)'
```

2015-12-22

ORBCOMM-2 (RTLS), 5 years after the first Falcon 9 flight

Q6 Boosters with drone-ship success and 4,000 < payload < 6,000 kg

```
SELECT Booster_Version
FROM SPACEXTABLE
WHERE Landing_Outcome = 'Success (drone ship)'
  AND PAYLOAD_MASS__KG_ > 4000
  AND PAYLOAD_MASS__KG_ < 6000
```

4 boosters

F9 FT B1022, F9 FT B1026, F9 FT B1021.2, F9 FT B1031.2

All Falcon 9 FT; two (.2) were reused boosters

EDA with SQL: Mission Outcomes, Max Payload and 2015 Failures

Q7 Total number of successful and failure mission outcomes

```
SELECT Mission_Outcome, COUNT(*) AS Total
FROM SPACEXTABLE
GROUP BY Mission_Outcome
```

Mission_Outcome	Total
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

Q8 Boosters that carried the maximum payload mass (sub-query)

```
SELECT Booster_Version
FROM SPACEXTABLE
WHERE PAYLOAD_MASS__KG_ =
  (SELECT MAX(PAYLOAD_MASS__KG_)
FROM SPACEXTABLE)
```

12 boosters · 15,600 kg

F9 B5 B1048.4, F9 B5 B1049.4, F9 B5 B1051.3, F9 B5 B1056.4, F9 B5 B1048.5, F9 B5 B1051.4, F9 B5 B1049.5, F9 B5 B1060.2, F9 B5 B1058.3, F9 B5 B1051.6, F9 B5 B1060.3, F9 B5 B1049.7

All Block 5 boosters, many reused

Q9 Failed drone-ship landings in 2015: month, booster version, launch site

```
SELECT substr(Date,6,2) AS Month,
  Landing_Outcome, Booster_Version, Launch_Site
FROM SPACEXTABLE
WHERE substr(Date,1,4)='2015'
  AND Landing_Outcome = 'Failure (drone ship)'
```

Month	Landing_Outcome	Booster_Version	Launch_Site
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

Key insight 100 of 101 missions succeeded: the risk is in landing the booster, not in reaching orbit.

EDA with SQL: Rank Landing Outcomes (2010-06-04 to 2017-03-20)

Q10 **Rank the count of landing outcomes between 2010-06-04 and 2017-03-20, in descending order**

```
SELECT Landing_Outcome, COUNT(*) AS Count
FROM SPACEXTABLE
WHERE Date BETWEEN '2010-06-04'

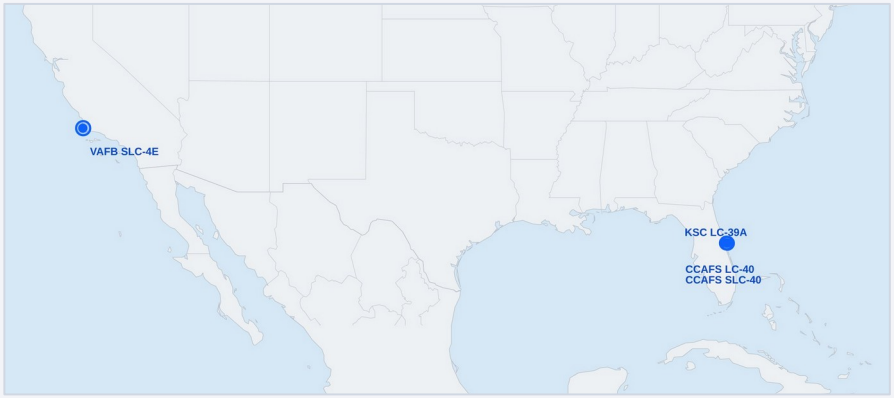
  AND '2017-03-20'
GROUP BY Landing_Outcome
ORDER BY Count DESC
```

Landing_Outcome	Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

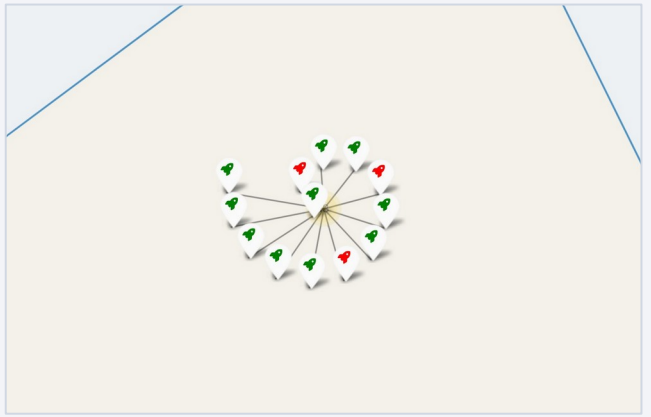


Key insight Before mid-2017, “No attempt” was the most common outcome (10); drone-ship successes and failures were tied (5 vs. 5).

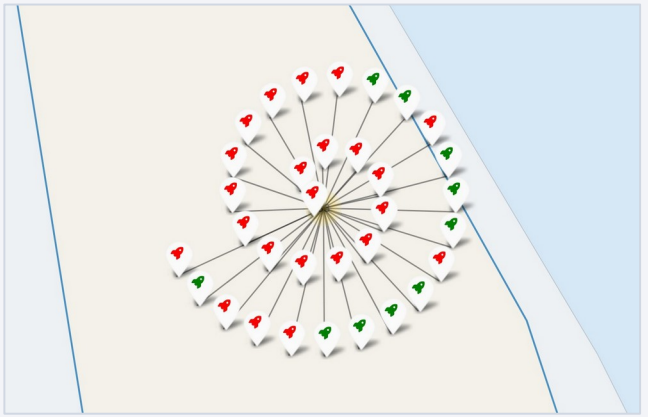
Folium Map Results: Launch Sites and Colour-Labelled Outcomes



All 4 launch-site markers



KSC LC-39A: 10 green / 13

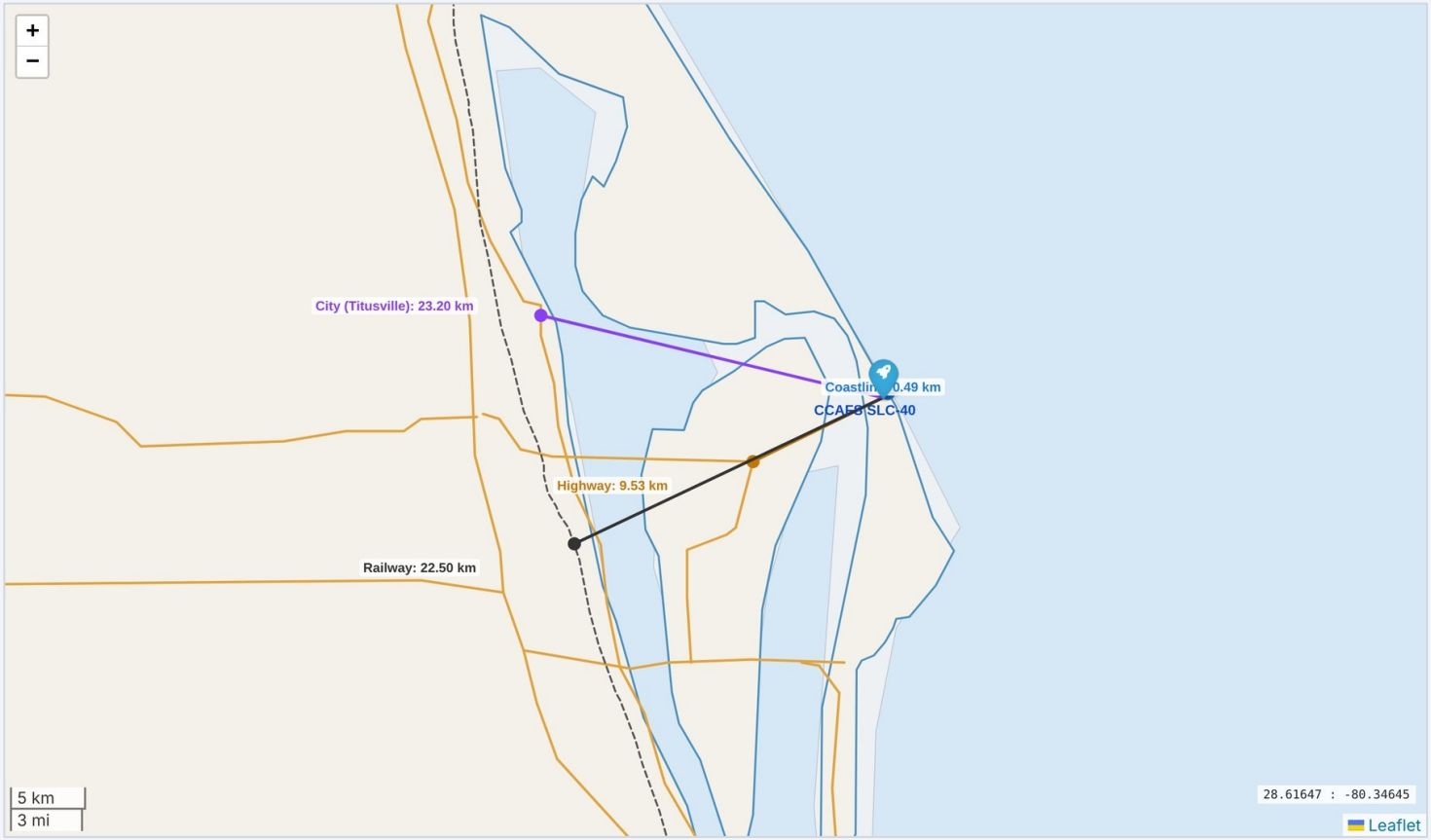


CCAFS LC-40 + SLC-40: 10 green / 33

- All launch sites are on the coast (Florida ~28.5°N, California ~34.6°N), so rockets fly over the ocean and drone-ship recovery is possible
- Green = landed, red = failed. Success ratio: KSC LC-39A 77%, CCAFS SLC-40 43%, VAFB SLC-4E 40%, CCAFS LC-40 27%

Key insight KSC LC-39A has the best success ratio; CCAFS LC-40 has the most failures because it hosted the early learning-phase flights.

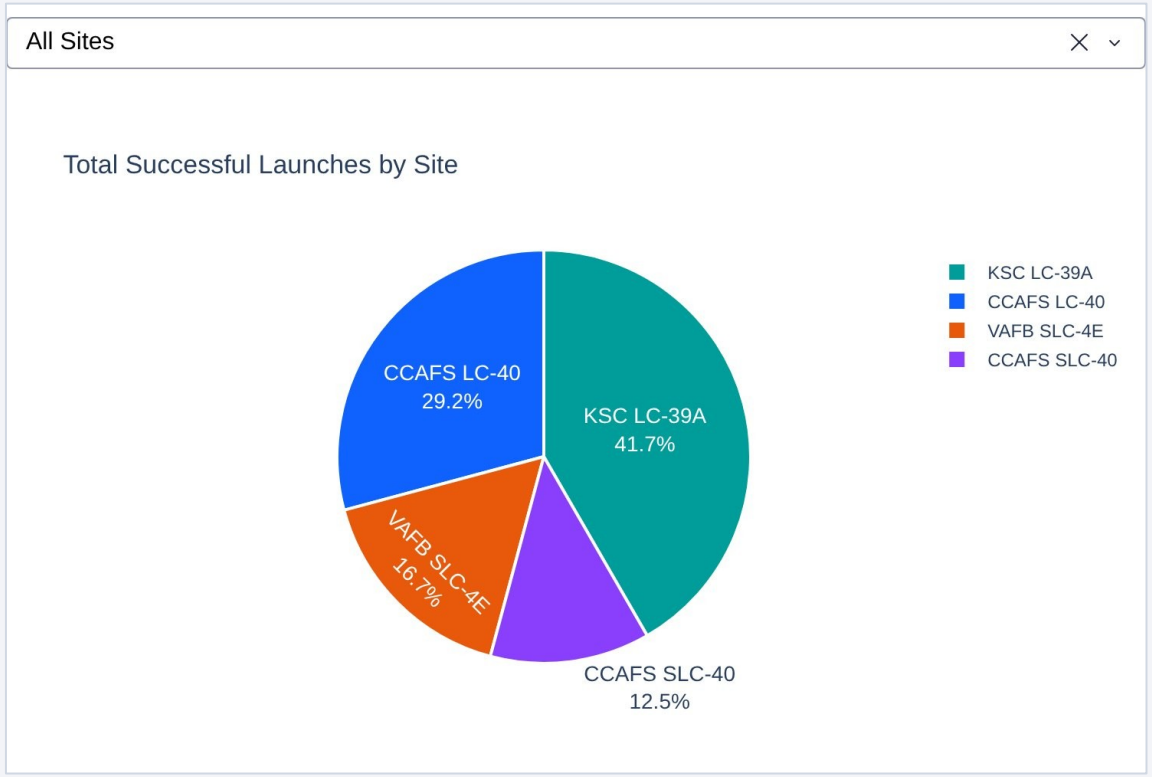
Folium Map Results: Proximity of CCAFS SLC-40 to Coast, Roads, Rail and City



Feature	Distance
Coastline	0.49 km
Highway (SR-405)	9.53 km
Railway (mainline)	22.50 km
City (Titusville)	23.20 km

- Very close to the coast: safe abort and splash-down over the ocean
- Kept away from cities, highways and mainline rail for public safety
- On-base roads and rail spurs are not in the 10 m basemap, so real access is closer

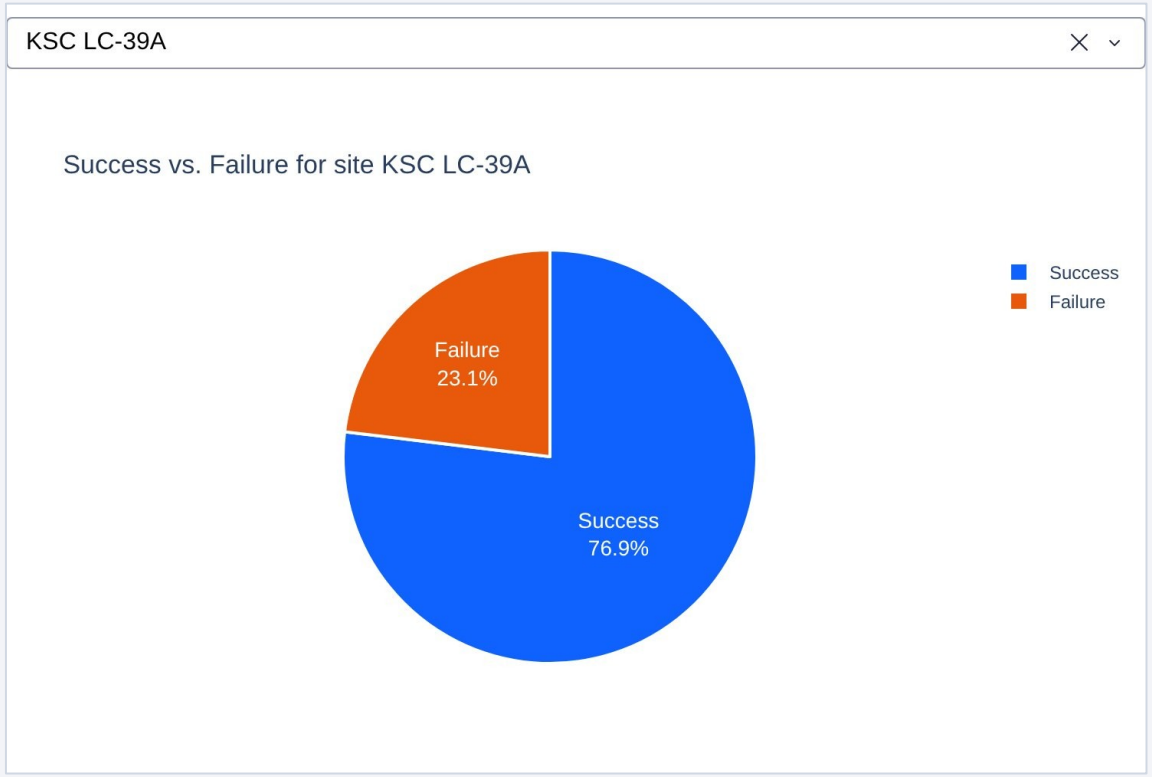
Plotly Dash Results: Launch Success Count for All Sites (Pie Chart)



- Dropdown = All Sites; the pie chart shows each site's share of all successful launches
  - KSC LC-39A 41.7%
  - CCAFS LC-40 29.2%
  - VAFB SLC-4E 16.7%
  - CCAFS SLC-40 12.5%

Key insight KSC LC-39A has the most successful launches of all sites, even though CCAFS LC-40 flew more launches (26 vs. 13).

Plotly Dash Results: KSC LC-39A, the Site with the Highest Success Ratio



76.9%

success ratio at KSC LC-39A
(10 of 13 launches)

- Pie chart for the selected site: 76.9% success vs. 23.1% failure (3 launches)
- Other sites: CCAFS SLC-40 42.9%, VAFB SLC-4E 40.0%, CCAFS LC-40 26.9%

Key insight KSC LC-39A leads both in share of successes and in success ratio; it was used only after the early learning phase.

Plotly Dash Results: Payload vs. Launch Outcome for All Sites (Range Slider)



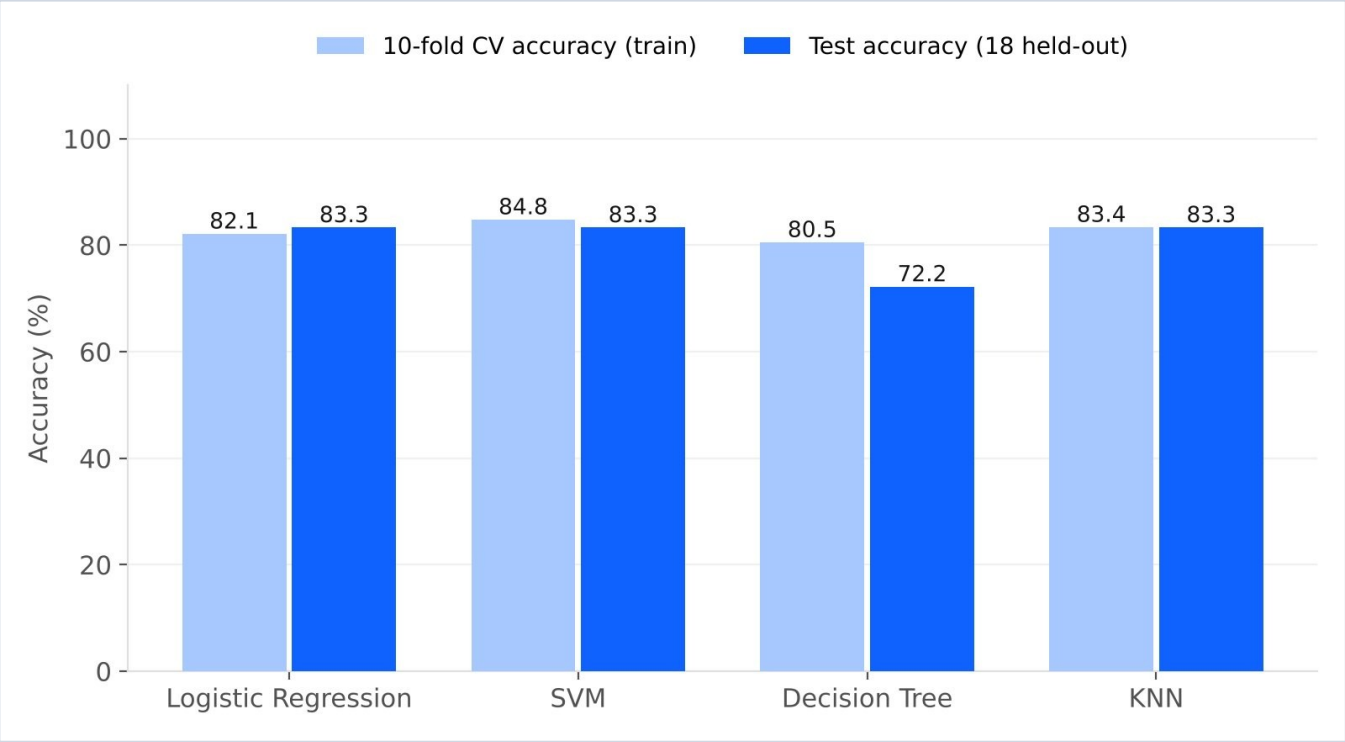
Slider 0–5,000 kg: 19 of 42 landed (45%)



Slider 5,000–10,000 kg: 3 of 11 landed (27%)

Key insight 2,000–4,000 kg has the highest success rate (62%); 6,000–8,000 kg never landed. Booster FT (67%) and B4 (55%) perform best; v1.0 (0%) and v1.1 (7%) almost never landed.

Predictive Analysis Results: Classification Accuracy

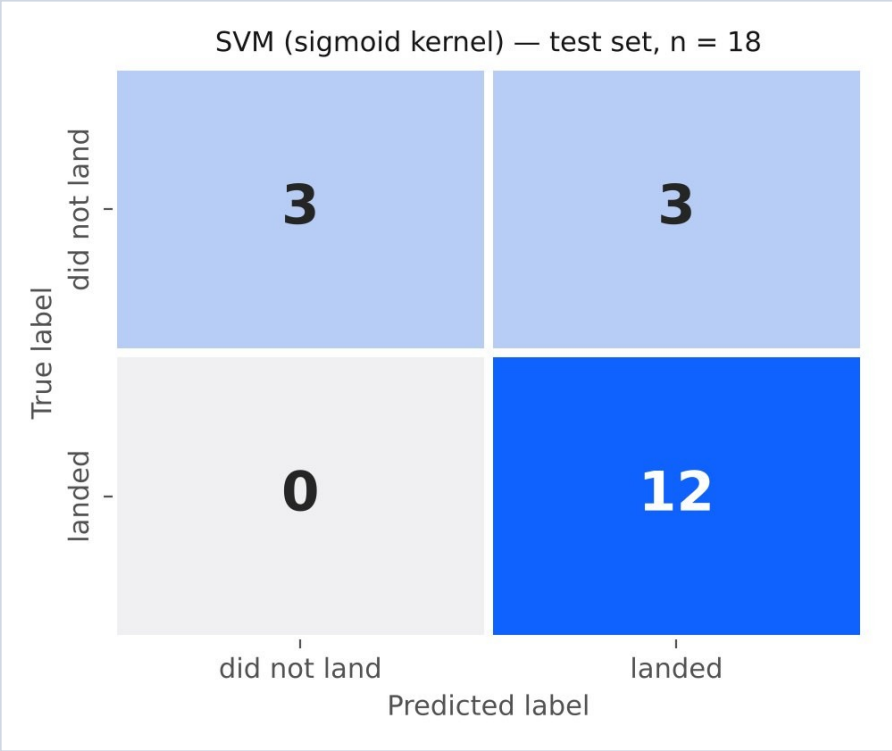


Model	CV acc.	Test acc.	F1
Logistic Regression	82.1%	83.3%	0.89
SVM	84.8%	83.3%	0.89
Decision Tree	80.5%	72.2%	0.80
KNN	83.4%	83.3%	0.89

- LogReg, SVM and KNN tie at 83.3% test accuracy (15 of 18); Decision Tree 72.2%
- Highest accuracy: SVM, with the best CV score (84.8%)
- Repeated CV: SVM 84.8% ± 7.3% stays best (baseline 66.7%)

Key insight Best model: SVM (sigmoid kernel, C = 1, γ = 0.032): 83.3% test accuracy and 84.8% CV accuracy.

Predictive Analysis Results: Confusion Matrix of the Best Model (SVM)



12
true positives

3
true negatives

3
false positives

- Recall = 100%: every real landing is predicted correctly (0 false negatives)
- Precision = 80%: the only error is predicting a landing for 3 boosters that were lost
- LogReg and KNN give the same matrix; the Decision Tree also misses 2 landings

Key insight False positives are the costly error for a bidder: they make SpaceX look cheaper than it is.

Conclusions

Experience drives success

Landing success rose from 0% (2010–13) to 84–90% (2019–20); flights 61–90 landed 83% of the time.

Hardware beats geography

Legs, grid fins, Block 5 and booster reuse have the strongest associations with landing.

Sites & orbits

KSC LC-39A has the best success ratio (77%). ES-L1, GEO, HEO, SSO were 100% but rare; GTO is hardest (52%).

Payload is secondary

2,000–4,000 kg is the sweet spot; heavy LEO/VLEO payloads land well, GTO is uncertain at any mass.

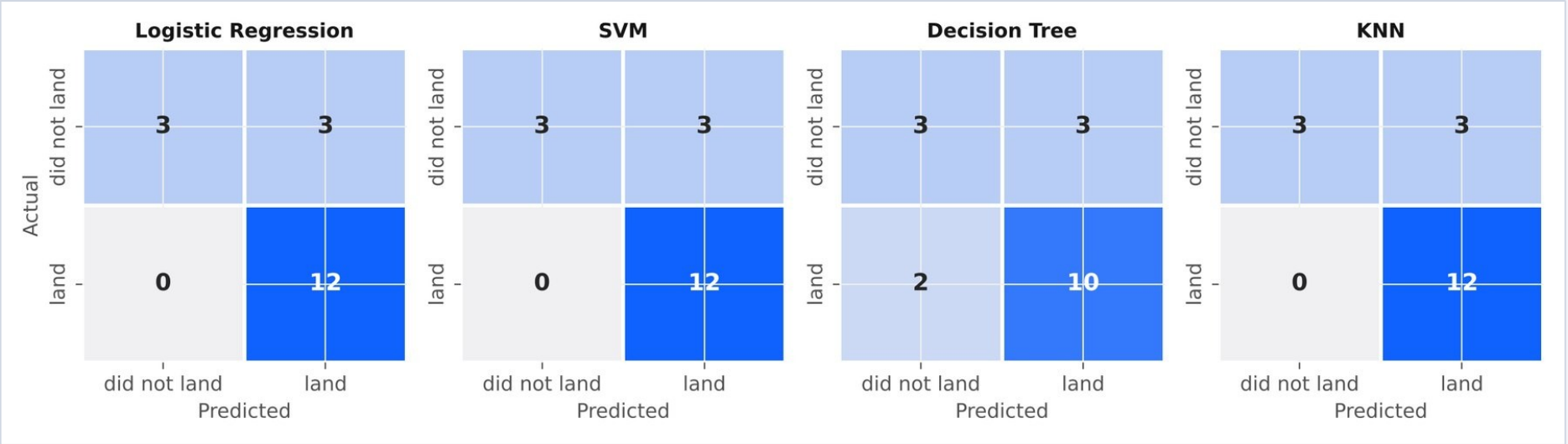
Prediction works

SVM is the best model: 83.3% test and 84.8% CV accuracy, with no missed landings.

Next steps

Add weather and booster-age features, weight recent flights, and retrain on post-2020 launches.

Appendix



- Confusion matrices of all 4 models on the 18-launch test set; top features: FlightNumber, PayloadMass, GridFins, ReusedCount
- Data: dataset_part_1/2/3.csv, Spacex.csv (101 rows), spacex_launch_geo.csv / spacex_launch_dash.csv (56 rows)
- Environment: Python 3, pandas, scikit-learn, seaborn, folium, dash/plotly, SQLite; split random_state = 2
- Folium basemap: Natural Earth 10 m layers; distances are Haversine to the nearest mapped feature

GitHub repository (all notebooks + Dash app): https://github.com/martimramos/ibmds